



East West University
Department of Computer Science and Engineering
Course: CSE103 (Structured Programming) LAB

LAB 03: Loops in C

Lab Objectives

- Understand how loops automate repetitive tasks in C.
- Learn the syntax and usage of for, while, and do-while loops.
- Use nested loops to generate patterns and handle multi-level iterations.
- Strengthen logic building through step-wise repetition and embedded iteration.

Topics Covered

- Basic Loops: for, while, do-while
- Loop initialization, condition, increment/decrement
- Counters and accumulators
- Nested Loops: loops inside loops
- Pattern generation and tabular outputs
- Loop control statements (break, continue)

Lab Outcomes

After completing this lab, students will be able to:

- Implement loops to repeat computations.
- Distinguish between for, while, and do-while.
- Apply nested loops to build patterns and tables.
- Trace loop execution and prevent infinite iterations.

Lab Activities

Part A: Basic Loops

Problem 1: Print Natural Numbers

Write a C program to print numbers from 1 to n using a for, while and do-while loop.

Input : 5	Output: 1 2 3 4 5
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Problem 2: Sum of First N Natural Numbers

Write a C program to sum numbers from 1 to n using a for loop and print the sum.

Input : 5	Output: Sum is 15
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Problem 3: Print letters

Write a C program that will take two integer numbers as input (x, y), where $1 \leq x$, $y \leq 26$ and $x \leq y$, and print the x-th letter to y-th letter in ascending order in English alphabet (uppercase only).

Sample Input 1 3	Sample Output A B C
Sample Input 3 7	Sample Output C D E F G

Problem 4: Sum of pairs

Write a C program that takes an integer number (n) as input and then it takes n pairs of integers numbers [(a1, b1), (a2, b2), (an, bn)] as input and displays the sum of each pair of numbers.

Sample input: 3 1 5 5 5 7 8	Sample Output: 6 10 15
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Problem 5: Multiple input

Write a C program that will take a pair of integer numbers (a and b) as input and will display the sum of those numbers. Your program should run until -1 is encountered.

Sample Input 2 3 5 10 10 10 -1	Sample Output 5 15 20
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Problem 6: Factorial of a Number

Input: 6	Output: Factorial = 720
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Problem 7: Reverse a Number

Input: 1234	Output: 4321
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Problem 8: Count Digits in a Number

Input: 10546	Output: 5
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Problem 9: Prime Number Check

Input: 17	Output: Prime Number
Input: 20	Output: Not a Prime Number

Problem 10 – Continuous Sum until 0 (do-while)

Keep taking numbers as input until the user enters 0.

Input: 5 10 15 0	Output: Sum = 30
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Part B: Nested Loops**Problem 1 : Star Pattern using nested loop**

Right Triangle Pattern	Inverted Right Triangle	Left-Aligned Triangle	Square Box Pattern
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Full Pyramid Pattern	Inverted Full Pyramid	Hollow Square Pattern	Hollow Pyramid Pattern
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Diamond Pattern	Hollow diamond Pattern	Rhombus Pattern	Butterfly Star Pattern
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Problem 2 : Number Pattern Programs

Number Triangle	Reverse Triangle	Increasing Pyramid	Pascal's Triangle
1 1 2 1 2 3 1 2 3 4 1 2 3 4 5	1 2 3 4 5 1 2 3 4 1 2 3 1 2 1	1 1 2 1 2 3 1 2 3 4 1 2 3 4 5	1 1 1 1 2 1 1 3 3 1 1 4 6 4 1
Number Changing	Number Triangular	Palindrome Number	Zero-One Triangle
1 2 3 4 5 6 7 8 9 10	1 2 2 3 3 3 4 4 4 4	1 1 2 1 1 2 3 2 1 1 2 3 4 3 2 1	1 0 1 1 0 1 0 1 0 1

Additional Practice & Resources

To strengthen your understanding of loops and nested loop, practice more examples and exercises from the following trusted learning resource:

Online Resource:

Link: <https://www.w3resource.com/c-programming-exercises/while-loop/index.php>

Link: <https://codeforwin.org/c-programming/for-do-while-loop-programming-exercises>
(Problem 1-26)